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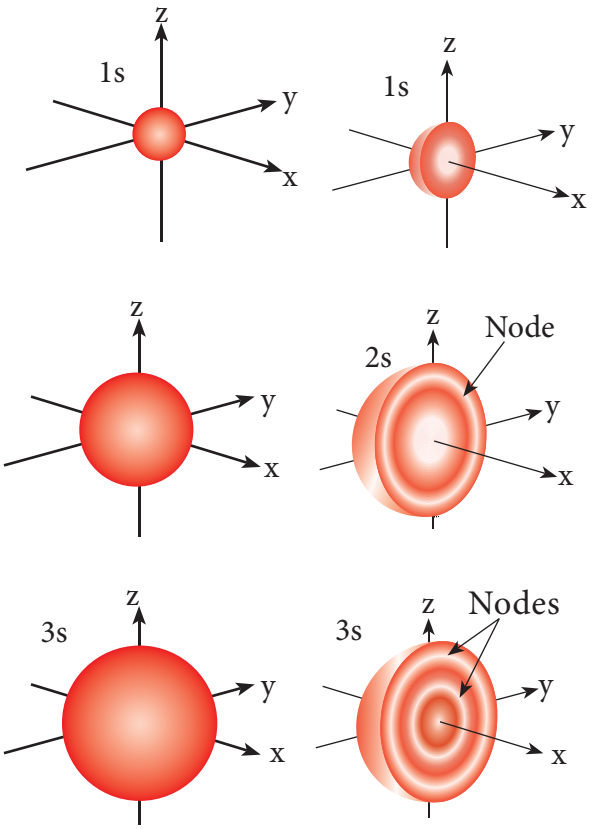
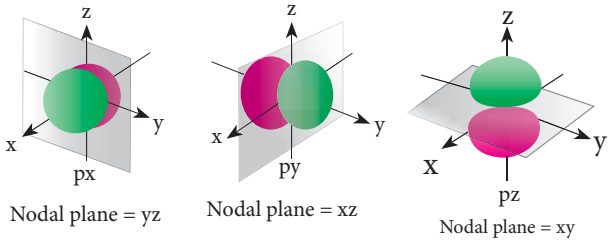


Figure 2.7 Shapes of 1s, 2s and 3s orbitals

For p orbitals l=1 and the corresponding m values are -1, 0 and +1. The angular distribution functions are quite complex and are not discussed here. The shape of the p orbital is shown in Figure 2.8. The three different m values indicates that there are three different orientations possible for p orbitals. These orbitals are designated as p_x, p_y and p_z and the angular distribution for these orbitals shows that the lobes are along the x, y and z axis respectively. As seen in the Figure 2.8 the 2p orbitals have one nodal plane.



(b) Cartoon representations of 2p orbitals

Figure 2.8 Shapes of 2p orbitals

For 'd' orbital l = 2 and the corresponding m values are -2, -1, 0 +1,+2. The shape of the d orbital looks like a 'clover leaf'. The five m values give rise to five d orbitals namely d_{xy} , d_{yz}, d_{zx}, $d_{x^2-y^2}$ and d_{z^2}. The 3d orbitals contain two nodal planes as shown in Figure 2.9.

